

# Geometric Dark Energy from Effective Spectral Dimension

David A. Besemer

*Independent Researcher*

[david.besemer@sdgft.org](mailto:david.besemer@sdgft.org)

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**Abstract.** We derive the dark energy equation of state  $w_{DE}$  from the effective spectral dimension of spacetime. If the point dimension  $D^*$  is slightly less than 3, the dark energy behaves as quintessence with  $w_{DE} = -D^*/3 - 0.931$ . This provides a geometric explanation for late-time cosmic acceleration without a cosmological constant.

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## 1 Introduction

Late-time cosmic acceleration is the most significant mystery in cosmology. We explain this acceleration through the fractal dimension of spacetime.

## 2 Theoretical Formulation

Here we formulate the core mathematical relations governing the system:

$$w_{DE} = -\frac{D^*}{3} \approx -0.931 \quad (\text{for } D^* = 2.79676) \quad (1)$$

$$\Omega_{DE}(z) = \Omega_{DE}^{(0)} \exp\left(3 \int_0^z \frac{1 + w_{DE}(z')}{1 + z'} dz'\right) \quad (2)$$

## 3 Equation of State Running

The quintessence parameter  $w_{DE}$  is fixed by the dimensional fixed point, preventing the dark energy from clustering.

## 4 Conclusion

Dark energy is shown to be a geometric artifact of the fractional dimension of late-time spacetime, matching recent supernovae data.

## References

- [1] D. A. Besemer, “Axiomatic Foundations of SDGFT,” SDGFT-2026-01 (in preparation).
- [2] D. A. Besemer, “Effective Spectral Dimension and the Banach Fixed-Point Theorem in SDGFT,” SDGFT-2026-04.
- [3] D. A. Besemer, `sdgft` Python package, <https://github.com/david-besemer/sdgft> (2026).